BCA Semester 1 Examination 2025

Model Solutions: 1BCA2 - PC Packages

Session	Paper Code	Time	Full Marks	Pass Marks
2024-2027	111602	3 Hours	100	40

Instructions: Answer any five questions (equal marks). Solutions for all nine questions are given below for practice. Menu names follow Microsoft Office 2016 and later.

Q1. What is macro? How is a macro created?

Macro

A macro is a recorded or programmed series of commands and actions that is saved under a name and can be run again with one click or a shortcut key. It automates repetitive tasks (formatting, data entry, report preparation), saves time and reduces errors. Macros are stored as VBA (Visual Basic for Applications) code and are available in Word, Excel and PowerPoint. Files with macros are saved as .xlsm (Excel) or .docm (Word).

Creating a macro by recording (Excel / Word)

1. Enable the Developer tab: File > Options > Customize Ribbon > tick Developer.
2. Click View > Macros > **Record Macro** (or Developer > Record Macro).
3. Enter a **Macro name** (no spaces), choose a **shortcut key** (e.g. Ctrl+Shift+M), choose where to store it (This Workbook / Personal Macro Workbook) and add a description. Click OK.
4. Perform the actions to be automated (for example set font to Arial 12, bold, add borders). Every step is recorded.
5. Click **Stop Recording**.

Running, editing and deleting

- Run: View > Macros > View Macros > select the macro > **Run**, or press the shortcut key, or assign it to a button.
- Edit: View Macros > Edit opens the VBA editor (Alt+F11) where the code can be changed; a macro can also be written directly in VBA.
- Delete: View Macros > select > Delete.
- Security: macros may contain viruses, so enable them only from trusted files (Trust Center > Macro Settings).

Q2. What is function and formula in MS Excel? Discuss some functions and formulas. How is a nested function used? Explain with example.

Formula

A formula is an expression entered by the user that starts with **=** and calculates a value from constants, cell references and operators (+ - * / ^). Examples: **=A1+B1**, **=(B2-C2)*10%**, **=A1^2**.

Function

A function is a ready-made formula built into Excel that performs a specific calculation. Syntax: **=NAME(arguments)**. It saves time and reduces mistakes. Excel has hundreds of functions in categories

such as Math, Statistical, Text, Date and Time, Logical and Lookup.

Function / formula	Purpose	Example
SUM	Total of numbers	=SUM(B2:B10)
AVERAGE	Mean value	=AVERAGE(B2:B10)
MAX / MIN	Highest / lowest value	=MAX(B2:B10)
COUNT / COUNTA	Count numeric / non-empty cells	=COUNT(A1:A20)
IF	Decision making	=IF(B2>=40,"Pass","Fail")
ROUND	Round a number	=ROUND(A1,2)
LEN, UPPER, LEFT, CONCAT	Text handling	=UPPER(A1)
TODAY, NOW, DATE	Date and time	=TODAY()
VLOOKUP	Find a value in a table	=VLOOKUP(101,A2:C20,3,FALSE)

Nested function

A nested function is a function used as an argument inside another function. Excel allows up to 64 levels of nesting. The inner function is evaluated first and its result is passed to the outer function.

Example 1: grade of a student using nested IF (marks in B2)

```
=IF(B2>=75,"Distinction",IF(B2>=60,"First",IF(B2>=40,"Pass","Fail")))
```

```
B2 = 82 -> Distinction      B2 = 65 -> First      B2 = 35 -> Fail
```

Example 2: `=ROUND(AVERAGE(B2:B6),1)` first finds the average of B2:B6 (e.g. 72.46) and ROUND then gives 72.5. **Example 3:** `=IF(AND(B2>=40,C2>=40),"Pass","Fail")` uses AND inside IF to test two subjects together.

Q3. Describe 10 different GUI components.

A Graphical User Interface (GUI) lets users interact with a computer using visual elements and a pointing device. Ten important components:

No.	Component	Description
1	Desktop	Main working screen that shows wallpaper, icons and shortcuts.
2	Icon	Small picture representing a file, folder, program or device; double-click opens it.
3	Window	Rectangular area that displays a running program or document; can be moved, resized, minimised, maximised or closed.
4	Menu	List of commands: menu bar, drop-down, pop-up (right-click) and Start menu.
5	Pointer / cursor	Arrow or hand moved by mouse/touchpad to select items.
6	Taskbar	Bar with the Start button, running applications, system tray and clock.
7	Toolbar / ribbon	Row of buttons for quick access to frequently used commands.
8	Dialog box	Small window that asks for input or confirmation, such as Save As or Print.
9	Scroll bar	Moves hidden parts of the content up/down or left/right.
10	Controls / widgets	Buttons, check boxes, radio buttons, text boxes, list and combo boxes, sliders and tabs used for input and choice.

Q4. What is disk defragmentation? Also discuss formatting of hard disk.

Disk defragmentation

When files are repeatedly created, edited and deleted, a file is stored in many non-adjacent pieces (fragments) over the disk. The read/write head must then jump around, which slows down the computer. **Defragmentation** is the process of rearranging these fragments so that each file is stored in contiguous clusters and free space is combined into one block.

- **Benefits:** faster file access and start-up, less head movement and wear, more contiguous free space.
- **Steps in Windows:** Open This PC, right-click the drive, Properties > Tools > **Optimize** (Defragment and Optimize Drives) > select drive > Analyze > Optimize.
- SSDs do not need defragmentation (Windows only runs TRIM on them), and it may reduce their life.

Formatting a hard disk

Formatting prepares a disk (or partition) for storing data by creating a file system and an empty directory structure; it normally erases the existing data.

- **Low-level format:** done by the manufacturer; creates tracks and sectors.
- **Partitioning:** dividing the disk into logical drives (C:, D:).
- **High-level format:** installs the file system such as FAT32, exFAT or NTFS, boot sector and file table. It can be *Quick* (removes the file table only) or *Full* (also checks for bad sectors).

```
Steps: This PC > right-click drive > Format > choose File system (NTFS),
       Allocation unit size, Volume label > tick Quick Format > Start
DOS command: FORMAT D: /FS:NTFS /Q
```

Warning: take a backup before formatting, because all data on the drive is lost.

Q5. Explain the process of the following features: (a) Mail Merge (b) Pivot table (c) Goal seek (d) Page setup

(a) Mail Merge (MS Word)

Mail Merge combines one main document (letter with merge fields) with a data source (Excel/Access list) to produce many personalised letters, labels or envelopes.

1 Type the letter in a new document.

2 Mailings > Start Mail Merge > Letters (or Labels/Envelopes/E-mail).

3 Mailings > Select Recipients > Use an Existing List and choose the data file.

4 Place the cursor and click Insert Merge Field to add <<Name>>, <<Address>> etc.

5 Click Preview Results; then Finish & Merge > Edit Individual Documents or Print Documents.

(b) Pivot table (MS Excel)

A pivot table summarises and analyses large data by grouping it and showing totals, counts or averages, and lets you rearrange rows and columns.

1 Select any cell in the data list (with column headings).

2 Insert > PivotTable > choose New Worksheet > OK.

3 Drag fields into Rows, Columns, Values and Filters areas of the PivotTable Fields pane.

4 Change the Value Field Settings (Sum, Count, Average) and refresh when data changes (Right-click > Refresh).

(c) Goal Seek (MS Excel)

Goal Seek finds the input value that makes a formula give a required result.

1 Enter the formula in a cell (e.g. B4 = average of marks).

2 Data > What-If Analysis > Goal Seek.

3 Set cell: B4; To value: 65; By changing cell: the input cell (e.g. B3).

4 Click OK; Excel shows the value needed. Click OK to keep it.

(d) Page Setup (Word/Excel)

Page Setup controls the layout of the printed page. Open Layout (Page Layout) tab > Page Setup dialog launcher.

- **Margins:** top, bottom, left, right and gutter.
- **Orientation:** Portrait or Landscape.
- **Size:** A4, Letter, Legal etc.
- **Layout/Header and Footer:** different first page, odd/even pages.
- In Excel also: Print Area, Print Titles, Scale to fit and gridlines.

Note: Pivot table and Goal Seek belong to MS Excel (the paper lists them under Word).

Q6. Explain the role of operating system in memory management.

Memory management is the function of the OS that controls and coordinates main memory (RAM), deciding which process gets memory, how much, and for how long.

Roles / functions

1 **Tracking:** keeps a record of which memory locations are free and which are allocated, and to which process.

2 **Allocation and deallocation:** allocates memory when a process is loaded and reclaims it when the process ends.

3 **Protection:** prevents a process from reading or writing the memory of another process or of the OS (base and limit registers).

4 **Swapping:** moves processes between main memory and disk to make room for others.

5 **Virtual memory:** uses part of the disk as an extension of RAM so programs larger than RAM can run.

6 **Relocation and sharing:** programs can be loaded at any address, and common code (libraries) can be shared.

Techniques

Technique	Idea
Contiguous allocation	Each process gets one continuous block (fixed or variable partitions); causes fragmentation.
Paging	Memory divided into fixed-size frames and the program into pages; pages can be loaded anywhere; page table maps them.

Technique	Idea
Segmentation	Program divided into logical segments (code, data, stack) of different sizes.
Demand paging / virtual memory	A page is loaded only when needed; page replacement (FIFO, LRU) when memory is full.

Fragmentation: external (free memory scattered in small holes) and internal (unused space inside an allocated block) reduce efficiency; compaction and paging help.

Q7. What are the various types of charts available in MS-Excel? Explain their uses.

Charts show worksheet data graphically (Insert > Charts). Select the data first, then choose the chart.

Chart	Use
Column	Compare values of different categories (vertical bars), e.g. monthly sales.
Bar	Same as column but horizontal; good for long category names.
Line	Show trends over time, e.g. temperature or share price.
Pie	Show each part as a share of the whole; best for one series with few items.
Doughnut	Like pie but with a hole; can show more than one series.
Area	Show how the total and its parts change over time.
XY (Scatter)	Show the relationship between two numeric variables.
Bubble	Scatter chart with a third value shown by bubble size.
Stock	Show open-high-low-close share prices.
Radar	Compare several variables of items from a centre point.
Surface	3-D view of the best combination of two sets of values.
Combo	Combines chart types, e.g. column for sales and line for profit %.
Histogram, Waterfall, Funnel, Treemap, Sunburst, Box and Whisker	Specialised charts for frequency, cumulative change, process stages and hierarchies.

Q8. Write steps for inserting watermark, header and footer, hyperlink.

Watermark (MS Word)

1. Open the document and go to the **Design** tab (Page Layout in older versions).
2. Click **Watermark** in the Page Background group.
3. Choose a built-in one (CONFIDENTIAL, DRAFT) or click **Custom Watermark**.
4. Select Text watermark (type text, font, size, colour, layout) or Picture watermark, click Apply and OK.
5. To remove: Design > Watermark > Remove Watermark.

Header and Footer

1. Go to the **Insert** tab and click **Header** (or Footer) in the Header & Footer group.
2. Choose a built-in style or Edit Header.

3 Type the text (title, name, date) in the header area; use the Header & Footer tab to add Page Number, Date and Time or a Picture.

4 Use Go to Footer to edit the footer in the same way.

5 Click **Close Header and Footer** (or press Esc). Double-clicking the header area also opens it.

Hyperlink

1 Select the text or object that should become a link.

2 Go to **Insert > Link** (Hyperlink), or press **Ctrl + K**.

3 In the dialog choose the link type: Existing File or Web Page (type the address), Place in This Document, Create New Document or E-mail Address.

4 Type the address (e.g. https://www.example.com), optionally a ScreenTip, and click OK.

5 Ctrl + click on the link to follow it; right-click > Remove Hyperlink to delete it.

Q9. Write the steps for: (i) Removing duplicate data (ii) Filter data (iii) Freezing pane (iv) Page orientation

(i) Removing duplicate data (Excel)

1 Select the range or click any cell in the table.

2 Go to **Data > Remove Duplicates**.

3 Tick the columns to compare (and 'My data has headers').

4 Click OK; Excel deletes duplicate rows and reports how many were removed.

(ii) Filter data

1 Click any cell in the data list with headings.

2 Data > **Filter** (Ctrl + Shift + L); arrows appear in the headings.

3 Click an arrow, tick the required values or use Text/Number/Date Filters (Equals, Greater Than, Between, Custom).

4 Click OK; only matching rows are shown. Data > Clear removes the filter.

(iii) Freezing panes

1 Select the cell just **below the rows** and **right of the columns** to be frozen (e.g. B2 freezes row 1 and column A).

2 Go to **View > Freeze Panes** and choose Freeze Panes (or Freeze Top Row / Freeze First Column).

3 To release: View > Freeze Panes > Unfreeze Panes.

(iv) Page orientation

1 Go to the **Layout** (Page Layout) tab.

2 Click **Orientation** in the Page Setup group.

3 Choose **Portrait** (vertical) or **Landscape** (horizontal). In Word, this can be applied to the whole document or only to a selected section.